EMV®
Integrated Circuit Card
Specifications for Payment Systems

Book 4

Cardholder, Attendant, and Acquirer Interface Requirements

Version 4.4
October 2022

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Revision Log – Version 4.4

The following changes have been made to Book 4 since the publication of Version 4.3. Numbering and cross references in this version have been updated to reflect changes introduced by the published bulletins.

Incorporated changes described in the following Specification Bulletins:

Specification Bulletin no. 103 : Unpredictable Number generation

Specification Bulletin no. 144: Terminal Unpredictable Number generation

Specification Bulletin no. 148: Clarification on Terminal Support of Multiple Application Version Numbers per AID

Specification Bulletin no. 151: Clarification on Cardholder Selection and Cardholder Confirmation

Specification Bulletin no. 163: Changes to PIN Pad requirements

Specification Bulletin no. 164: Electronic Signature Capture and Electronic Receipt Delivery

Specification Bulletin no. 178, Third Edition: Tokenisation Data Objects – Payment Account Reference (PAR)

Specification Bulletin no. 185, Second Edition: Biometric Terminal Specification

Specification Bulletin 197: Tokenisation Data Objects – Token Requestor ID and Last 4 Digits of PAN

Specification Bulletin no. 220, Second Edition: EMV® Contact Kernel Output

Specification Bulletin no. 231: Issuer Identification Number Extended (IINE)

Specification Bulletin no. 243: Introduction of XDA/ODE for EMV Specifications

Minor editorial clarifications and corrections, including those described in the following:

Specification Bulletin no. 243: Introduction of XDA/ODE for EMV Specifications

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Part I

General

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1 Scope

This document, the *Integrated Circuit Card Specifications for Payment Systems - Book 4, Cardholder, Attendant, and Acquirer Interface Requirements for Payment Systems*, defines the mandatory, recommended, and optional terminal requirements necessary to support the acceptance of integrated circuit cards (ICCs) in accordance with the other documents of the *Integrated Circuit Card Specifications for Payment Systems*, all available on http://www.emvco.com:

- Book 1 – Application Independent ICC to Terminal Interface Requirements
- Book 2 – Security and Key Management
- Book 3 – Application Specification

1.1 Changes in Version 4.4

This release incorporates all relevant Specification Bulletins, Application Notes, amendments, etc. published up to the date of this release.

The Revision Log at the beginning of the Book provides additional detail about changes to this specification.

1.2 Structure

Book 4 consists of the following parts:

Part I	-	**General**
Part II	-	**General Requirements**
Part III	-	**Software Architecture**
Part IV	-	**Cardholder, Attendant, and Acquirer Interface**
Part V	-	**Annexes**

Part I includes this introduction, as well as data applicable to all Books: normative references, definitions, abbreviations, notations, data element format convention, and terminology.

Part II addresses:

- Functional requirements, such as those emerging from the other Books of the *Integrated Circuit Card Specifications for Payment Systems*
- General physical characteristics

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Part III addresses software architecture including software and data management.

Part IV discusses:

- Cardholder and attendant interface
- Acquirer interface

Part V discusses the coding of terminal data elements, lists the common character set, provides an example data element conversion, includes informative terminal guidelines, and provides examples of the physical and functional characteristics of terminals.

The Book also includes a revision log and an index.

This specification applies to all terminals operating in attended or unattended environments, having offline or online capabilities, and supporting transaction types such as purchase of goods, services, and cash. Terminals include but are not limited to automated teller machines (ATMs), branch terminals, cardholder-activated terminals, electronic cash registers, personal computers, and point of service (POS) terminals.

This specification defines the requirements necessary to support the implementation of ICCs. These requirements are in addition to those already defined by individual payment systems and acquirers for terminals that accept magnetic stripe cards. ICC and magnetic stripe acceptance capability may co-exist in the same terminal.

It is recognised that different terminal implementations exist depending on business environment and intended usage. This specification defines requirements for those features and functions that are applicable according to the particular operating environment of the terminal.

This specification:

- Does not cover application-specific terminal requirements unique to individual payment systems and those functions not required to support interchange.
- Does not address cardholder or merchant operating procedures, which are established by individual payment systems.
- Does not provide sufficient detail to be used as a specification for terminal procurement.

Individual payment systems and acquirers will define complementary requirements applicable to different situations that will provide more detailed specifications applicable to terminal implementations.

1.3 Underlying Standards

This specification is based on the ISO/IEC 7816 series of standards and should be read in conjunction with those standards. However, if any of the provisions or definitions in this specification differ from those standards, the provisions herein shall take precedence.

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1.4 Audience

This specification is intended for use by manufacturers of ICCs and terminals, system designers in payment systems, and financial institution staff responsible for implementing financial applications in ICCs.

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2 Normative References

The following specifications and standards contain provisions that are referenced in these specifications. The latest version shall apply unless a publication date is explicitly stated.

EMV Contact Interface Specification	EMV Level 1 Specifications for Payment Systems, EMV Contact Interface Specification
EMV Tokenisation Framework	EMV Payment Tokenisation Specification – Technical Framework Framework specification for an interoperable Payment Tokenisation solution.
FIPS 202	SHA-3 Standard: Permutation-Based Hash and Extendable-Output Functions
IEEE P1363	Standard Specifications For Public-Key Cryptography
ISO 639-1	Codes for the representation of names of languages – Part 1: Alpha-2 Code **Note:** This standard is updated continuously by ISO. Additions/changes to ISO 639-1:1988: Codes for the Representation of Names of Languages are available on: http://www.loc.gov/standards/iso639-2/php/code_changes.php
ISO 3166	Codes for the representation of names of countries and their subdivisions
ISO 4217	Codes for the representation of currencies and funds
ISO/IEC 7812-1	Identification cards – Identification of issuers — Part 1: Numbering System
ISO/IEC 7813	Identification cards – Financial transaction cards
ISO/IEC 7816-4	Identification cards — Integrated circuit cards — Part 4: Organization, security and commands for interchange
ISO/IEC 7816-5	Identification cards — Integrated circuit cards — Part 5: Registration of application providers
ISO/IEC 7816-6	Identification cards – Integrated circuit cards – Part 6: Interindustry data elements for interchange

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ISO/IEC 7816-11	Identification cards – Integrated circuit cards – Personal verification through biometric methods
ISO 8583:1987	Bank card originated messages – Interchange message specifications – Content for financial transactions
ISO 8583:1993	Financial transaction card originated messages – Interchange message specifications
ISO/IEC 8825-1	Information technology – ASN.1 encoding rules: Specification of Basic Encoding Rules (BER), Canonical Encoding Rules (CER) and Distinguished Encoding Rules (DER)
ISO/IEC 8859	Information processing – 8-bit single-byte coded graphic character sets
ISO 9362	Banking – Banking telecommunication messages – Bank identifier codes
ISO 9564-1	Financial services – Personal Identification Number (PIN) management and security – Part 1: Basic principles and requirements for PINs in card-based systems
ISO/IEC 9796-2	Information technology – Security techniques – Digital signature schemes giving message recovery – Part 2: Integer factorization based mechanisms
ISO/IEC 9797-1	Information technology – Security techniques – Message Authentication Codes – Part 1: Mechanisms using a block cipher
ISO/IEC 9797-2	Information technology – Security techniques – Message Authentication Codes (MACs) – Part 2: Mechanisms using a dedicated hash-function
ISO/IEC 10116	Information technology – Security techniques – Modes of operation for an n-bit block cipher
ISO/IEC 10118-3	Information technology – Security techniques – Hash-functions – Part 3: Dedicated hash-functions
ISO/IEC 11770-6	Information technology – Security techniques – Key management — Part 6: Key derivation
ISO 13616	Banking and related financial services – International bank account number (IBAN)

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ISO/IEC 14888-3	Information technology – Security techniques – Digital signatures with appendix — Part 3: Discrete logarithm based mechanisms
ISO/IEC 15946-1	Information technology – Security techniques – Cryptographic techniques based on elliptic curves — Part 1: General
ISO/IEC 15946-5	Information technology – Security techniques – Cryptographic techniques based on elliptic curves — Part 5: Elliptic curve generation
ISO 16609	Banking – Requirements for message authentication using symmetric techniques
ISO/IEC 18031	Information technology – Security techniques – Random bit generation
ISO/IEC 18033-2	Information technology – Security techniques – Encryption algorithms – Part 2: Asymmetric ciphers
ISO/IEC 18033-3	Information technology – Security techniques – Encryption algorithms – Part 3: Block ciphers
ISO/IEC 19772	Information technology – Security techniques – Authenticated encryption
ISO/IEC 19785-3	Information technology – Common Biometric Exchange Formats Framework – Patron format specifications
ISO/IEC 19794	Information technology – Biometric data interchange formats
ISO/IEC 19794-2	Information technology – Biometric data interchange formats – Part 2: Finger minutiae data
SEC 1	Elliptic Curve Cryptography (available at http://www.secg.org)

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3 Definitions

The following terms are used in one or more books of these specifications.

Application	The application protocol between the card and the terminal and its related set of data.
Application Authentication Cryptogram	An Application Cryptogram generated by the card when declining a transaction.
Application Cryptogram	A cryptogram generated by the card in response to a GENERATE AC command. See also: • Application Authentication Cryptogram • Authorisation Request Cryptogram • Transaction Certificate
Authentication	The provision of assurance of the claimed identity of an entity or of data origin.
Authorisation Request Cryptogram	An Application Cryptogram generated by the card when requesting online authorisation.
Authorisation Response Cryptogram	A cryptogram generated by the issuer in response to an Authorisation Request Cryptogram.
Biometric Data Block	A block of data with a specific format that contains information captured from a biometric capture device and that could be used as follows: • stored in the card as part of the biometric reference template • sent to the ICC in the data field of the PIN CHANGE/UNBLOCK command • sent to the ICC in the data field of the VERIFY command for offline biometric verification • sent online for verification The format of the BDB is outside the scope of this specification.

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Biometric Reference Template	Biometric data stored in the card as reference. Data provided by a biometric capture device would be compared against the biometric reference template to determine a match.
Biometric Verification	The process of determining that the biometrics presented, such as finger, palm, iris, voice, or facial, are valid.
Byte	8 bits.
Card	A payment card as defined by a payment system.
Certificate	The public key and identity of an entity together with some other information, rendered unforgeable by signing with the private key of the certification authority which issued that certificate.
Certification Authority	Trusted third party that establishes a proof that links a public key and other relevant information to its owner.
Ciphertext	Enciphered information.
Combined DDA/Application Cryptogram Generation	A form of offline dynamic data authentication.
Command	A message sent by the terminal to the ICC that initiates an action and solicits a response from the ICC.
Command Chaining	A mechanism where consecutive command-response pairs can be chained.
Compromise	The breaching of secrecy or security.
Concatenation	Two elements are concatenated by appending the bytes from the second element to the end of the first. Bytes from each element are represented in the resulting string in the same sequence in which they were presented to the terminal by the ICC, that is, most significant byte first. Within each byte bits are ordered from most significant bit to least significant. A list of elements or objects may be concatenated by concatenating the first pair to form a new element, using that as the first element to concatenate with the next in the list, and so on.

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Contact	A conducting element ensuring galvanic continuity between integrated circuit(s) and external interfacing equipment.
Cryptogram	Result of a cryptographic operation.
Cryptographic Algorithm	An algorithm that transforms data in order to hide or reveal its information content.
Data Integrity	The property that data has not been altered or destroyed in an unauthorised manner.
Decipherment	The reversal of a corresponding encipherment.
DEM1	A family of data encapsulation mechanisms defined in ISO/IEC 18033-2.
Digital Signature	An asymmetric cryptographic transformation of data that allows the recipient of the data to prove the origin and integrity of the data, and protect the sender and the recipient of the data against forgery by third parties, and the sender against forgery by the recipient.
Dynamic Data Authentication	A form of offline dynamic data authentication
Elliptic Curve Cryptography	Public key cryptography based on the algebraic structure of elliptic curves over finite fields.
Encipherment	The reversible transformation of data by a cryptographic algorithm to produce ciphertext.
Exclusive-OR	Binary addition with no carry, giving the following values: 0 + 0 = 0 0 + 1 = 1 1 + 0 = 1 1 + 1 = 0
Extended Data Authentication	A form of offline dynamic data authentication.
Facial Verification	The process of determining that the face presented is valid.
Financial Transaction	The act between a cardholder and a merchant or acquirer that results in the exchange of goods or services against payment.

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Finger Verification	The process of determining that the finger presented is valid.
Function	A process accomplished by one or more commands and resultant actions that are used to perform all or part of a transaction.
Hash Function	A function that maps strings of bits to fixed-length strings of bits, satisfying the following two properties: • It is computationally infeasible to find for a given output an input which maps to this output. • It is computationally infeasible to find for a given input a second input that maps to the same output. Additionally, if the hash function is required to be collision-resistant, it must also satisfy the following property: • It is computationally infeasible to find any two distinct inputs that map to the same output.
Hash Result	The string of bits that is the output of a hash function.
I2OSP	An integer to octet string conversion primitive function defined in ISO/IEC 18033-2.
Integrated Circuit(s)	Electronic component(s) designed to perform processing and/or memory functions.
Integrated Circuit(s) Card	A card into which one or more integrated circuits are inserted to perform processing and memory functions.
Interface Device	That part of a terminal into which the ICC is inserted, including such mechanical and electrical devices as may be considered part of it.
Iris Verification	The process of determining that the iris presented is valid.
Issuer Action Code	Any of the following, which reflect the issuer-selected action to be taken upon analysis of the TVR: • Issuer Action Code – Default • Issuer Action Code – Denial • Issuer Action Code – Online

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Kernel	The set of functions required to be present on every terminal implementing a specific interpreter. The kernel contains device drivers, interface routines, security and control functions, and the software for translating from the virtual machine language to the language used by the real machine. In other words, the kernel is the implementation of the virtual machine on a specific real machine.
Key	A sequence of symbols that controls the operation of a cryptographic transformation.
Key Introduction	The process of generating, distributing, and beginning use of a key pair.
Key Withdrawal	The process of removing a key from service as part of its revocation.
Keypad	Arrangement of numeric, command, and, where required, function and/or alphanumeric keys laid out in a specific manner.
Library	A set of high-level software functions with a published interface, providing general support for terminal programs and/or applications.
Logical Compromise	The compromise of a key through application of improved cryptanalytic techniques, increases in computing power, or combination of the two.
Magnetic Stripe	The stripe containing magnetically encoded information.
Message	A string of bytes sent by the terminal to the card or vice versa, excluding transmission-control characters.
Message Authentication Code	A symmetric cryptographic transformation of data that protects the sender and the recipient of the data against forgery by third parties.
Nibble	The four most significant or least significant bits of a byte.
Offline Data Encipherment	Offline encipherment of data, in particular for cardholder PIN and biometric data. See Book 2.
Padding	Appending extra bits to either side of a data string.
Palm Verification	The process of determining that the palm presented is valid.

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Path	Concatenation of file identifiers without delimitation.
Payment System Environment	A logical construct within the ICC, the entry point to which is a Directory Definition File (DDF) named '1PAY.SYS.DDF01'. This DDF contains a Payment System Directory which in turn contains entries for one or more Application Definition Files (ADFs) which are formatted according to this specification.
Physical Compromise	The compromise of a key resulting from the fact that it has not been securely guarded, or a hardware security module has been stolen or accessed by unauthorised persons.
PIN Pad	Arrangement of numeric and command keys to be used for personal identification number (PIN) entry. Also known as a "PIN Entry Device" (PED).
Plaintext	Unenciphered information.
Potential Compromise	A condition where cryptanalytic techniques and/or computing power has advanced to the point that compromise of a key of a certain length is feasible or even likely.
Private Key	That key of an entity's asymmetric key pair that should only be used by that entity. In the case of a digital signature scheme, the private key defines the signature function.
Public Key	That key of an entity's asymmetric key pair that can be made public. In the case of a digital signature scheme, the public key defines the verification function.
Public Key Certificate	The public key information of an entity signed by the certification authority and thereby rendered unforgeable.
Response	A message returned by the ICC to the terminal after the processing of a command message received by the ICC.
RSA-KEM	A family of key encapsulation mechanisms defined in ISO/IEC 18033-2.
RSATransform	The RSA exponentiation that is used for encryption and decryption, and generating and verifying a signature.

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Script	A command or a string of commands transmitted by the issuer to the terminal for the purpose of being sent serially to the ICC as commands.
Secret Key	A key used with symmetric cryptographic techniques and usable only by a set of specified entities.
Socket	An execution vector defined at a particular point in an application and assigned a unique number for reference.
Static Data Authentication	Offline static data authentication
Symmetric Cryptographic Technique	A cryptographic technique that uses the same secret key for both the originator's and recipient's transformation. Without knowledge of the secret key, it is computationally infeasible to compute either the originator's or the recipient's transformation.
Template	Value field of a constructed data object, defined to give a logical grouping of data objects.
Terminal	The device used in conjunction with the ICC at the point of transaction to perform a financial transaction. The terminal incorporates the interface device and may also include other components and interfaces such as host communications.
Terminal Action Code	Any of the following, which reflect the acquirer-selected action to be taken upon analysis of the TVR: • Terminal Action Code – Default • Terminal Action Code – Denial • Terminal Action Code – Online
Terminate Card Session	End the card session by deactivating the IFD contacts according to EMV Contact Interface Specification and displaying a message indicating that the ICC cannot be used to complete the transaction.
Terminate Transaction	Stop the current application and deactivate the card.
Transaction	An action taken by a terminal at the user's request. For a POS terminal, a transaction might be payment for goods, etc. A transaction selects among one or more applications as part of its processing flow.

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Transaction Certificate	An Application Cryptogram generated by the card when accepting a transaction.
Virtual Machine	A theoretical microprocessor architecture that forms the basis for writing application programs in a specific interpreter software implementation.
Voice Verification	The process of determining that the voice presented is valid.

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4 Abbreviations, Notations, Conventions, and Terminology

4.1 Abbreviations

a	Alphabetic (see section 4.3, Data Element Format Conventions)
AAC	Application Authentication Cryptogram
AAD	Additional Authenticated Data
AC	Application Cryptogram
ADF	Application Definition File
AEF	Application Elementary File
AES	Advanced Encryption Standard
AFL	Application File Locator
AID	Application Identifier
AIP	Application Interchange Profile
an	Alphanumeric (see section 4.3)
ans	Alphanumeric Special (see section 4.3)
APDU	Application Protocol Data Unit
API	Application Program Interface
ARC	Authorisation Response Code
ARPC	Authorisation Response Cryptogram
ARQC	Authorisation Request Cryptogram
ASI	Application Selection Indicator
ASN	Abstract Syntax Notation
ATC	Application Transaction Counter
ATM	Automated Teller Machine
ATR	Answer to Reset

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AUC	Application Usage Control
b	Binary (see section 4.3)
BCD	Binary Coded Decimal
BDB	Biometric Data Block
BEK	Biometric Encryption Key
BER	Basic Encoding Rules (defined in ISO/IEC 8825-1)
BHT	Biometric Header Template
BIC	Bank Identifier Code
BIT	Biometric Information Template
BMK	Biometric MAC Key
CA	Certification Authority
CAD	Card Accepting Device
C-APDU	Command APDU
CBC	Cipher Block Chaining
CBEFF	Common Biometric Exchange Formats Framework
CCD	Common Core Definitions
CCI	Common Core Identifier
CCYYMMDD	Year (4 digits), Month, Day
CDA	Combined DDA/Application Cryptogram Generation
CDOL	Card Risk Management Data Object List
CID	Cryptogram Information Data
CLA	Class Byte of the Command Message
cn	Compressed Numeric (see section 4.3)
CPU	Central Processing Unit
CRL	Certificate Revocation List
CSU	Card Status Update
CV	Cryptogram Version
CV Rule	Cardholder Verification Rule

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CVM	Cardholder Verification Method
CVR	Card Verification Results
DDA	Dynamic Data Authentication
DDF	Directory Definition File
DDOL	Dynamic Data Authentication Data Object List
DES	Data Encryption Standard
DF	Dedicated File
DIR	Directory
DOL	Data Object List
ECB	Electronic Code Book
EC-SDSA	Elliptic Curve Schnorr Digital Signature Algorithm
ECC	Elliptic Curve Cryptography
EF	Elementary File
EN	European Norm
FC	Format Code
FCI	File Control Information
Hex	Hexadecimal
HHMMSS	Hours, Minutes, Seconds
HMAC	Keyed-hash Message Authentication Code
I/O	Input/Output
IAC	Issuer Action Code (Denial, Default, Online)
IAD	Issuer Application Data
IBAN	International Bank Account Number
IC	Integrated Circuit
ICC	Integrated Circuit(s) Card
ICCD	Issuer Certified Card Data
IEC	International Electrotechnical Commission
IFD	Interface Device

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IIN	Issuer Identification Number
IINE	Issuer Identification Number Extended
INS	Instruction Byte of Command Message
ISO	International Organization for Standardization
KD	Key Derivation
KDF	Key Derivation Function
K_M	Master Key
K_S	Session Key
L	Length
l.s.	Least Significant
Lc	Exact Length of Data Sent by the TAL in a Case 3 or 4 Command
LCOL	Lower Consecutive Offline Limit
L_{DD}	Length of the ICC Dynamic Data
Le	Maximum Length of Data Expected by the TAL in Response to a Case 2 or 4 Command
Lr	Length of Response Data Field
LRC	Longitudinal Redundancy Check
M	Mandatory
m.s.	Most Significant
MAC	Message Authentication Code
max.	Maximum
MF	Master File
MK	ICC Master Key for session key generation
MMDD	Month, Day
MMYY	Month, Year
n	Numeric (see section 4.3)
N_{CA}	Length of the Certification Authority Public Key Modulus

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NF	Norme Française
N_{FIELD}	Length of a finite field element
N_{HASH}	Output length of a hash function
N_I	Length of the Issuer Public Key Modulus
N_{IC}	Length of the ICC Public Key Modulus
NIST	National Institute for Standards and Technology
N_{PE}	Length of the ICC PIN Encipherment Public Key Modulus
N_{SIG}	Length of an ECC Digital Signature
O	Optional
O/S	Operating System
ODA	Offline Data Authentication
ODE	Offline Data Encipherment
P1	Parameter 1
P2	Parameter 2
PAN	Primary Account Number
PAR	Payment Account Reference
PC	Personal Computer
P_{CA}	Certification Authority Public Key
PDOL	Processing Options Data Object List
P_I	Issuer Public Key
P_{IC}	ICC Public Key
PIN	Personal Identification Number
PIX	Proprietary Application Identifier Extension
POS	Point of Service
pos.	Position
PSE	Payment System Environment
R-APDU	Response APDU

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RFU	Reserved for Future Use
RID	Registered Application Provider Identifier
RSA	Rivest, Shamir, Adleman Algorithm
S_{CA}	Certification Authority Private Key
SDA	Static Data Authentication
SDAD	Signed Dynamic Application Data
SFI	Short File Identifier
SHA-1	Secure Hash Algorithm 1
SHA-2	Secure Hash Algorithm 2 (includes SHA-256 and SHA-512)
SHA-256	Secure Hash Algorithm 256
SHA-3	Secure Hash Algorithm 3
S_I	Issuer Private Key
S_{IC}	ICC Private Key
SK	Session Key
SW1	Status Byte One
SW2	Status Byte Two
TAA	Terminal Action Analysis
TAC	Terminal Action Code(s) (Default, Denial, Online)
TAL	Terminal Application Layer
TC	Transaction Certificate
TDOL	Transaction Certificate Data Object List
TLV	Tag Length Value
TPDU	Transport Protocol Data Unit
TSI	Transaction Status Information
TVR	Terminal Verification Results
UCOL	Upper Consecutive Offline Limit
UL	Underwriters Laboratories Incorporated
UN	Unpredictable Number

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var.	Variable (see section 4.3)
XDA	Extended Data Authentication
YYMM	Year, Month
YYMMDD	Year, Month, Day

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4.2 Notations

'0' to '9' and 'A' to 'F'	16 hexadecimal characters
xx	Any value
A := B	A is assigned the value of B
A = B	Value of A is equal to the value of B
A ≡ B mod n	Integers A and B are congruent modulo the integer n, that is, there exists an integer d such that $(A - B) = dn$
A mod n	The reduction of the integer A modulo the integer n, that is, the unique integer r, $0 \leq r < n$, for which there exists an integer d such that $A = dn + r$
A / n	The integer division of A by n, that is, the unique integer d for which there exists an integer r, $0 \leq r < n$, such that $A = dn + r$
Y := ALG(K)[X]	Encipherment of a data block X with a block cipher as specified in Book 2 section A1, using a secret key K
$X = ALG^{-1}(K)[Y]$	Decipherment of a data block Y with a block cipher as specified in Book 2 section A1, using a secret key K
$Y := Sign\ (S_K)[X]$	The signing of a data block X with an asymmetric reversible algorithm as specified in Book 2 section A2, using the private key S_K
$X = Recover(P_K)[Y]$	The recovery of the data block X with an asymmetric reversible algorithm as specified in Book 2 section A2, using the public key P_K
C := (A || B)	The concatenation of an n-bit number A and an m-bit number B, which is defined as $C = 2^m A + B$.
Leftmost	Applies to a sequence of bits, bytes, or digits and used interchangeably with the term "most significant". If C = (A || B) as above, then A is the leftmost n bits of C.

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Rightmost	Applies to a sequence of bits, bytes, or digits and used interchangeably with the term "least significant". If C = (A || B) as above, then B is the rightmost m bits of C.
H := Hash[MSG]	Hashing of a message MSG of arbitrary length using a 160-bit hash function
X ⊕ Y	The symbol '⊕' denotes bit-wise exclusive-OR and is defined as follows: X ⊕ Y — The bit-wise exclusive-OR of the data blocks X and Y. If one data block is shorter than the other, then it is first padded to the left with sufficient binary zeros to make it the same length as the other.
MIN (x, y)	The smaller of values x and y.

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4.3 Data Element Format Conventions

The EMV specifications use the following data element formats:

a	Alphabetic data elements contain a single character per byte. The permitted characters are alphabetic only (a to z and A to Z, upper and lower case).
an	Alphanumeric data elements contain a single character per byte. The permitted characters are alphabetic (a to z and A to Z, upper and lower case) and numeric (0 to 9). There is one exception: The permitted characters for Payment Account Reference are alphabetic *upper case* (A to Z) and numeric (0 to 9).
ans	Alphanumeric Special data elements contain a single character per byte. The permitted characters and their coding are shown in the Common Character Set table in Book 4 Annex B. There is one exception: The permitted characters for Application Preferred Name are the non-control characters defined in the ISO/IEC 8859 part designated in the Issuer Code Table Index associated with the Application Preferred Name.
b	These data elements consist of either unsigned binary numbers or bit combinations that are defined elsewhere in the specification. Binary example: The Application Transaction Counter (ATC) is defined as "b" with a length of two bytes. An ATC value of 19 is stored as Hex '00 13'. Bit combination example: Processing Options Data Object List (PDOL) is defined as "b" with the format shown in Book 3 section 5.4.
cn	Compressed numeric data elements consist of two numeric digits (having values in the range Hex '0'–'9') per byte. These data elements are left justified and padded with trailing hexadecimal 'F's. Example: The Application Primary Account Number (PAN) is defined as "cn" with a length of up to ten bytes. A value of 1234567890123 may be stored in the Application PAN as Hex '12 34 56 78 90 12 3F FF' with a length of 8.
n	Numeric data elements consist of two numeric digits (having values in the range Hex '0' – '9') per byte. These digits are right justified and padded with leading hexadecimal zeroes. Other specifications sometimes refer to this data format as Binary Coded Decimal ("BCD") or unsigned packed. Example: Amount, Authorised (Numeric) is defined as "n 12" with a length of six bytes. A value of 12345 is stored in Amount, Authorised (Numeric) as Hex '00 00 00 01 23 45'.
var.	Variable data elements are variable length and may contain any bit combination. Additional information on the formats of specific variable data elements is available elsewhere.

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4.4 Terminology

business agreement	An agreement reached between a payment system and its business partner(s).
proprietary	Not defined in this specification and/or outside the scope of this specification
shall	Denotes a mandatory requirement
should	Denotes a recommendation

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Part II

General Requirements

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5 Terminal Types and Capabilities

5.1 Terminal Types

As described in section 1, this specification addresses a broad spectrum of terminals. For the purpose of this specification, terminals are categorised by the following:

- Environment: Attended or unattended
- Communication: Online or offline
- Operational control: Financial institution, merchant, or cardholder

Table 1 defines the terms used to describe terminal types.

Term	Definition
Attended	An attendant (an agent of the merchant or of the acquirer) is present at the point of transaction and participates in the transaction by entering transaction-related data. The transaction occurs 'face to face'.
Unattended	The cardholder conducts the transaction at the point of transaction without the participation of an attendant (agent of the merchant or of the acquirer). The transaction does not occur 'face to face'.
Online only	The transaction can normally only be approved in real time by transmission of an authorisation request message.
Offline with online capability	Depending upon transaction characteristics, the transaction can be completed offline by the terminal or online in real time. It is equivalent to 'online with offline capability'.
Offline only	The transaction can only be completed offline by the terminal.
Operational control	Identifies the entity responsible for the operation of the terminal. This does not necessarily equate to the actual owner of the terminal.

Table 1: Terms Describing Terminal Types

Within this specification, *online* reflects online communication to acquirer (or its agent). The acquirer is assumed to be capable of communicating to the issuer (or its agent).

The type of terminal shall be indicated in Terminal Type. The coding of Terminal Type using the three categories is shown in Annex A.

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5.2 Terminal Capabilities

For the purpose of this specification, terminal capabilities are described in Terminal Capabilities and Additional Terminal Capabilities.

The following categories shall be indicated in Terminal Capabilities:

- **Card data input capability** - Indicates all the methods supported by the terminal for entering the information from the card into the terminal.
- **Cardholder Verification Method (CVM) capability** - Indicates all the methods supported by the terminal for verifying the identity of the cardholder at the terminal.
- **Security capability** - Indicates all the methods supported by the terminal for authenticating the card at the terminal and whether or not the terminal has the ability to capture a card.

The following categories shall be indicated in Additional Terminal Capabilities:

- **Transaction type capability** - Indicates all the types of transactions supported by the terminal.
- **Terminal data input capability** - Indicates all the methods supported by the terminal for entering transaction-related data into the terminal.
- **Terminal data output capability** - Indicates the ability of the terminal to print or display messages and the character set code table(s) referencing the part(s) of ISO/IEC 8859 supported by the terminal.

The coding of Terminal Capabilities and Additional Terminal Capabilities using these categories is shown in Annex A.

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5.3 Terminal Configurations

Terminal capabilities and device components vary depending on the intended usage and physical environment. A limited set of configuration examples follow.

Figure 1 illustrates an example of an attended terminal where the integrated circuit (IC) interface device (IFD) and PIN pad are integrated but separate from the POS device (such as for an electronic fund transfer terminal or an electronic cash register).

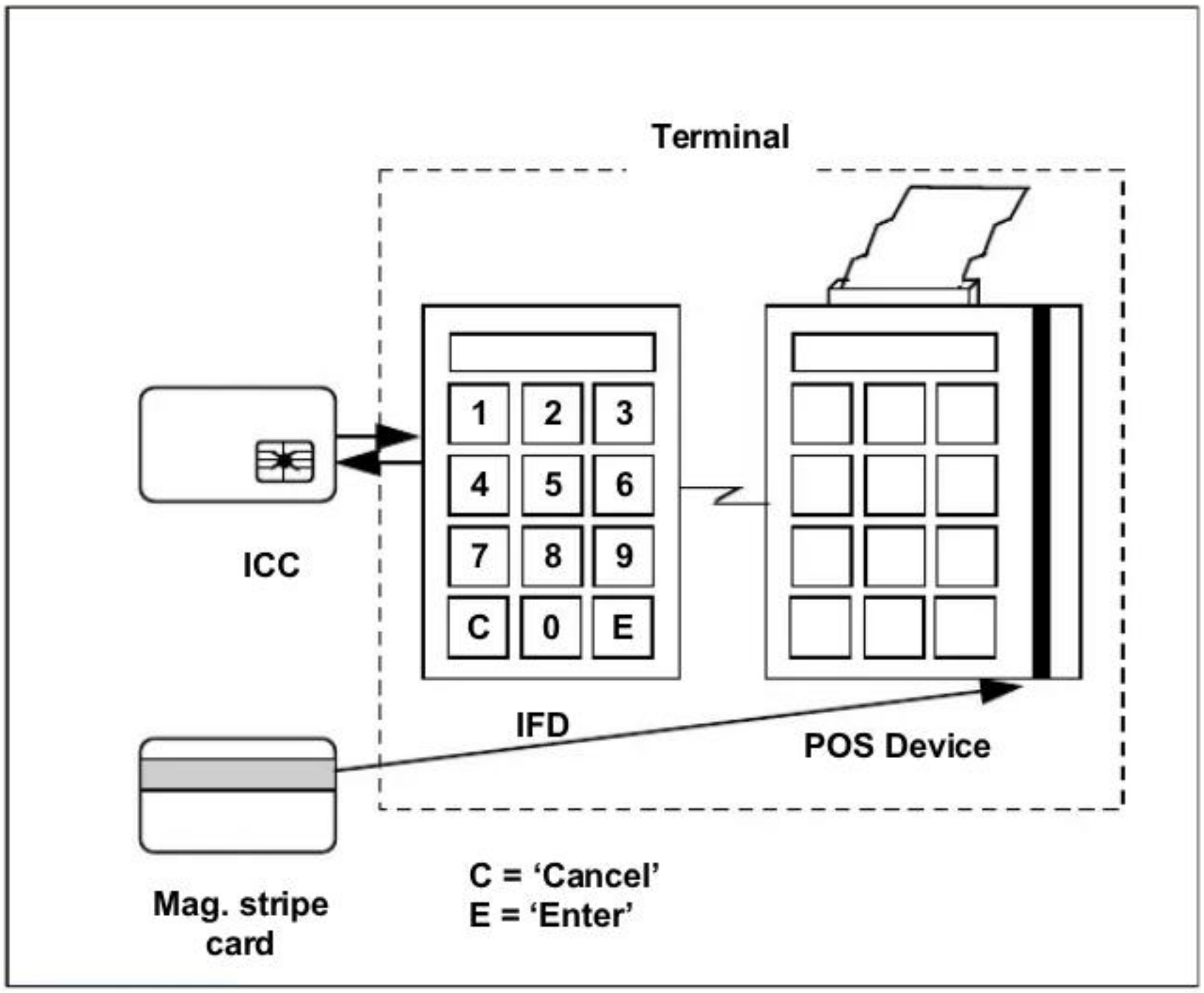


Figure 1: Example of an Attended Terminal

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Figure 2 illustrates an example of merchant host concentrating devices, which may be of various types and capabilities.

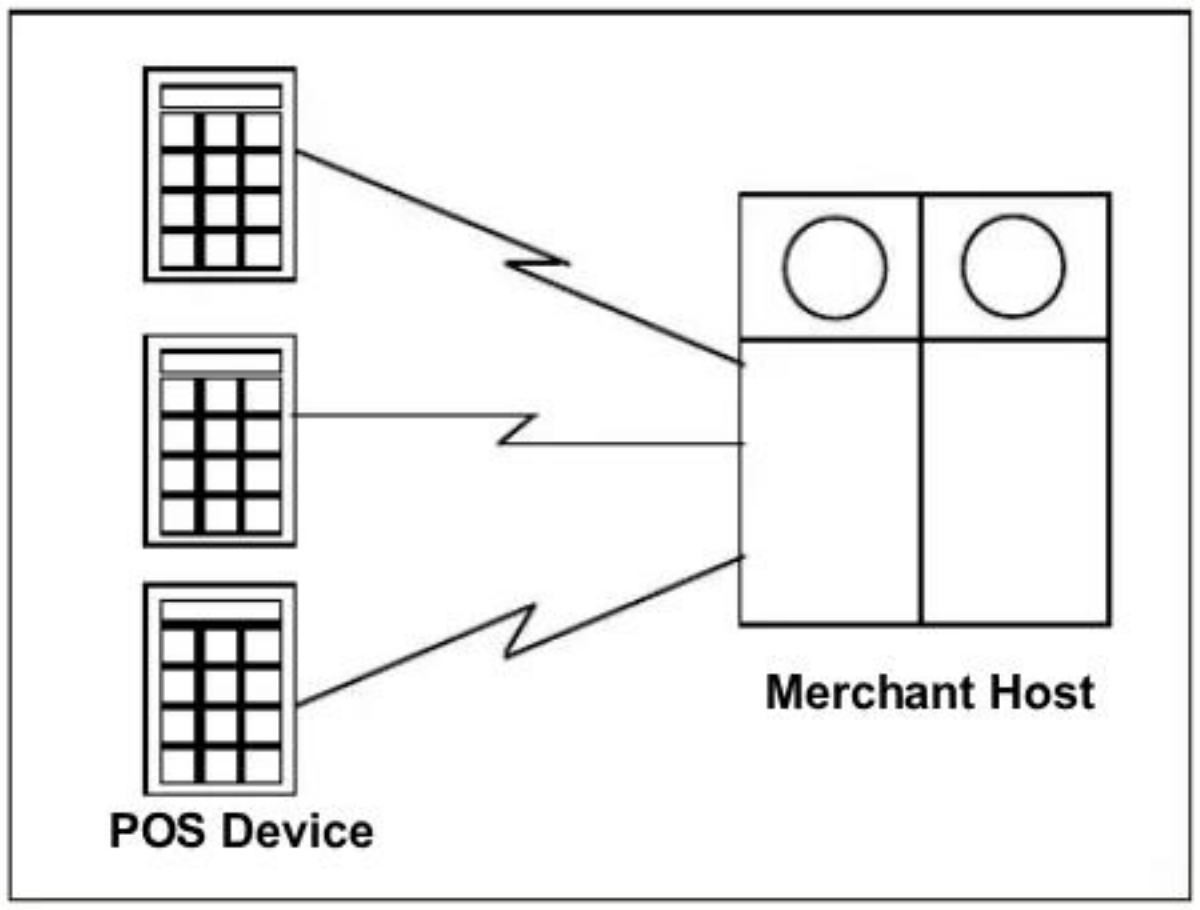


Figure 2: Example of a Merchant Host

Within this specification a merchant host to which is connected a cluster of POS devices shall be considered, in its totality, as a 'terminal' regardless of the distribution of functions between the host and POS devices. (See section 10 for terminal data management requirements.)

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Figure 3 illustrates an example of a cardholder-controlled terminal that is connected via a public network to a merchant or acquirer host.

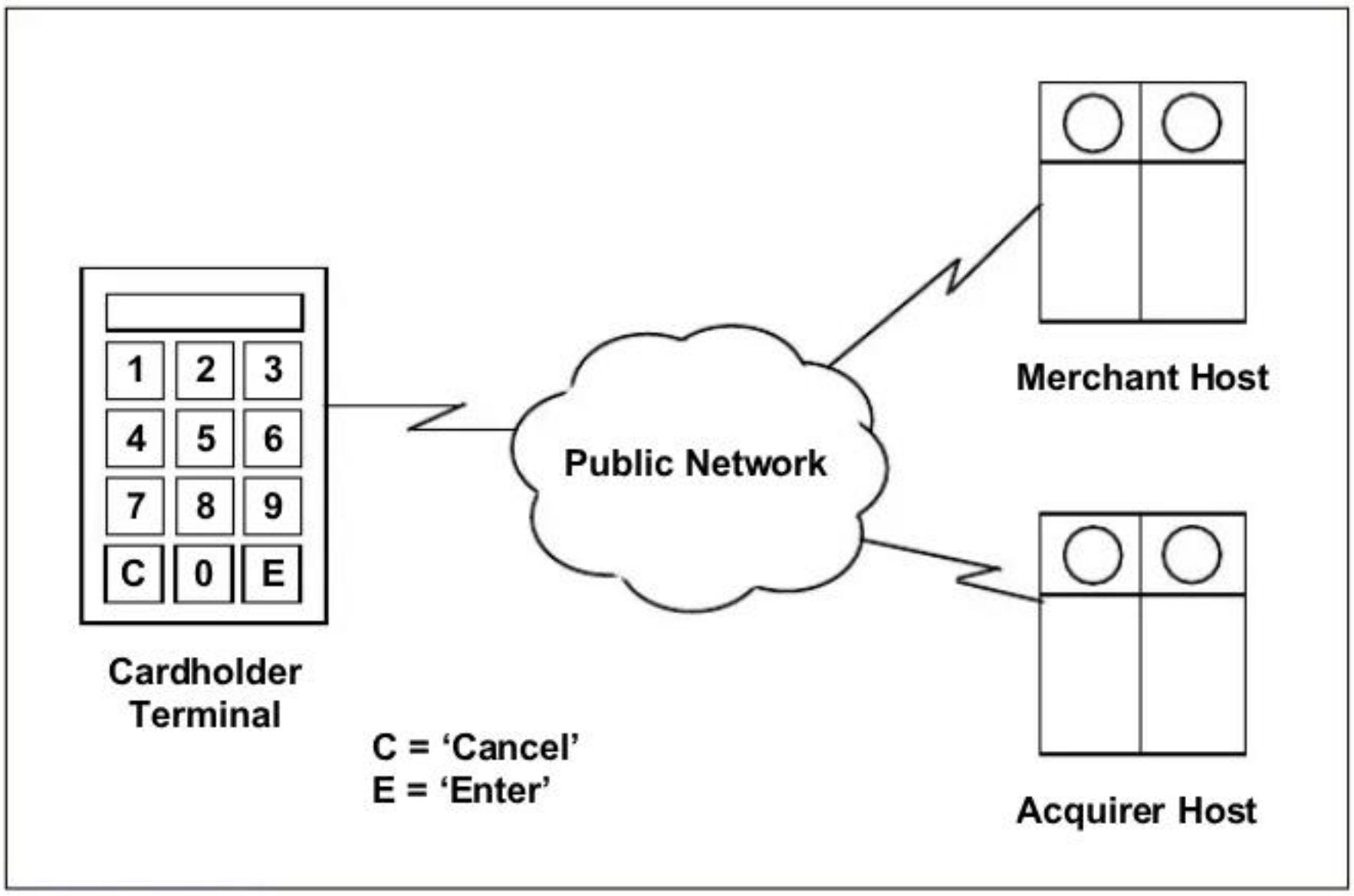


Figure 3: Example of a Cardholder-Controlled Terminal

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6 Functional Requirements

This Book does not replicate the other Books of the *Integrated Circuit Card Specifications for Payment Systems* but describes the implementation issues and the impact of those Books on the terminal.

This section uses standard messages described in section 11.2 to illustrate the appropriate message displays for the transaction events described below.

The usage of Authorisation Response Code, CVM Results, and Issuer Script Results is specified in this section. See Annex A for additional information on coding.

6.1 Application Independent ICC to Terminal Interface Requirements

The terminal shall comply with all Parts of Book 1. It shall support all data elements and commands subject to the conditions described in section 6.3.

6.2 Security and Key Management

The terminal shall comply with all Parts of Book 2. It shall support all data elements and commands subject to the conditions described in section 6.3.

6.3 Application Specification

The terminal shall comply with all Parts of Book 3. It shall support all functions subject to the conditions described in this section.

Sections 6.3.1 to 6.3.9 expand upon the terminal functions described in Book 3.

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6.3.1 Initiate Application Processing

When the Processing Options Data Object List (PDOL) includes an amount field (either Amount, Authorised or Amount, Other), an attended terminal (Terminal Type = 'x1', 'x2', or 'x3') shall provide the amount at this point in transaction processing. If the amount is not yet available, the terminal shall obtain the amount and should display the 'ENTER AMOUNT' message. For any other terminal type, if the terminal is unable to provide the amount at this point in transaction processing, the amount field in the data element list shall be filled with hexadecimal zeroes.

As described in Book 3, if the card returns SW1 SW2 = '6985' in response to the GET PROCESSING OPTIONS command, indicating that the transaction cannot be performed with this application, then the terminal should display the 'NOT ACCEPTED' message and shall return to application selection. The terminal shall not allow that application to be selected again for this card session as defined in EMV Contact Interface Specification.

6.3.2 Offline Data Authentication

An online-only terminal supporting no form of offline data authentication as indicated in Terminal Capabilities shall set to 1 the 'Offline data authentication was not performed' bit in the Terminal Verification Results (TVR). (For details, see Book 3 Annex C.)

All other terminals shall be capable of performing SDA, DDA, optionally CDA, and optionally XDA, as described in Books 2 and 3.

6.3.2.1 CDA

The following section applies when the selected form of offline data authentication is CDA.

When CDA fails prior to the final Terminal Action Analysis (for example, Issuer Public Key recovery fails) preceding the issuance of a first GENERATE AC command or a second GENERATE AC command in the case 'unable to go online', the terminal shall set the TVR bit for 'CDA failed' to 1 and request the cryptogram type determined by Terminal Action Analysis. In this case, the GENERATE AC command shall not request a CDA signature and no further CDA processing is performed.

When a CDA failure is detected after the final Terminal Action Analysis preceding the issuance of a first or second GENERATE AC command, the terminal shall set the 'CDA failed' bit in the TVR to 1 and the following rules apply:

If CDA fails in conjunction with the first GENERATE AC:

- If the Cryptogram Information Data (CID) indicates that the card has returned a TC, the terminal shall decline the transaction and not perform a second GENERATE AC command.
- If the CID indicates that the card has returned an ARQC, the terminal shall complete the transaction processing by performing an immediate second GENERATE AC command requesting an AAC.

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If CDA fails in conjunction with the second GENERATE AC, the terminal shall decline the transaction

If as part of dynamic signature verification the CID was retrieved from the ICC Dynamic Data (as recovered from the Signed Dynamic Application Data), then it is this value that shall be used to determine the cryptogram type. Otherwise the cleartext CID in the GENERATE AC response shall be used.

6.3.2.2 XDA

When the selected form of offline data authentication is XDA, the terminal shall perform the processing defined in the following sections when XDA fails. The first three sections address the three different ways in which XDA can fail and the fourth section describes terminal processing after first GENERATE AC XDA failure.

Note: In the following sections, 'final TAA' means the Terminal Action Analysis that occurs immediately before issuance of a GENERATE AC command.

6.3.2.2.1 CA ECC Public Key Retrieval Failure

If the terminal experiences a CA ECC Public Key retrieval error, the terminal shall set the 'CA ECC key missing' bit in the TVR to 1 *before* the final TAA preceding the first GENERATE AC command. In this case the terminal omits ECC key recovery and XDA signature verification and continues with the processing defined in section 6.3.2.2.4.

6.3.2.2.2 ECC Key Recovery Failure

If the terminal experiences an ECC Issuer Public Key or ICC Public Key recovery error, the terminal shall:

- Set the 'ECC key recovery failed' bit in the TVR *after* issuance of the first GENERATE AC command but *before* the TAA that occurs following the first GENERATE AC command. However, the 'ECC key recovery failed' bit is not set if the 'CA ECC key missing' bit is set.
- Perform the processing defined in section 6.3.2.2.4.

Note: The terminal may perform ECC key recovery before issuance of the first GENERATE AC command, but setting of the 'ECC key recovery failed' bit in the TVR must occur after issuance of the first GENERATE AC command to ensure that the result of final TAA preceding the first GENERATE AC command does not vary based upon when ECC key recovery is performed by the terminal. If the TVR is included as input to generation of the Application Cryptogram, the issuer should set the 'ECC key recovery failed' bit in the TVR to 0 when performing Application Cryptogram verification, as the value of that bit may have changed after the Application Cryptogram was generated.

6.3.2.2.3 XDA Signature Verification Failure

After a GENERATE AC command, if either of the following is true:

- the Signed Dynamic Application Data (SDAD) was not received in the GENERATE AC response
- verification of the Signed Dynamic Application Data (SDAD) failed

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then when ECC key recovery has ended, the terminal shall:

- Set the 'XDA signature verification failed' bit in the TVR to 1 if CA ECC key retrieval and ECC key recovery were successful.
- For the first GENERATE AC, perform the processing defined in section 6.3.2.2.4.
- For the second GENERATE AC, decline the transaction. **Note:** This will only apply if XDA was successful on the first GENERATE AC as otherwise section 6.3.2.2.4 applies which specifies that the XDA signature on the second GENERATE AC is not verified.

6.3.2.2.4 Terminal Processing After First GENERATE AC XDA Failure

The terminal performs the processing described in this section when it experiences an XDA failure, specifically a CA ECC public key retrieval failure (as defined in section 6.3.2.2.1), an ECC key recovery failure (as defined in section 6.3.2.2.2) or an XDA signature verification failure for first GENERATE AC (as defined in section 6.3.2.2.3).

If the CID indicated that the card returned a TC or ARQC, the terminal shall:

- Perform TAA using the TAC-Denial and IAC-Denial. If the IAC-Denial does not exist, a default value with all bits set to 0 is used.
- If the result of this TAA is to decline the transaction (that is, a TVR bit is 1 and the corresponding TAC-Denial or IAC-Denial bit is also 1), the terminal shall:
  - If the CID indicated that the card returned a TC, no second GENERATE AC command shall be issued to complete the transaction.
  - If the CID indicated that the card returned an ARQC, a second GENERATE AC command requesting an AAC shall be issued to complete the transaction.

Otherwise (the result of this TAA is not to decline the transaction), the terminal shall attempt to send the transaction online for authorisation[1] and:

- The TVR value included in the online authorisation shall include the updated values of the 'ECC key recovery failed' and the 'XDA signature verification failed' bits.

  Note: If the TVR is included as input to generation of the Application Cryptogram, the issuer should set the 'ECC key recovery failed' and the 'XDA signature verification failed' bits in the TVR to 0 when performing Application Cryptogram verification, as the value of those bits may have changed after the Application Cryptogram was generated.

- If the terminal is able to process the transaction online, then:
  - If the CID indicated that the card returned a TC, the terminal shall not issue a second GENERATE AC command to complete the transaction. Depending on the Authorisation Response Code returned in the response message, the terminal shall determine whether to accept or decline the transaction.

[1] In this scenario, a cryptogram type of TC may be sent in the online authorisation.

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- If the CID indicated that the card returned an ARQC, the terminal shall issue a second GENERATE AC command to complete the transaction as described in section 6.3.8. Although an XDA signature will be requested it shall not be verified by the terminal.[2]

- If the terminal is for any reason unable to process the transaction online, then:
  - The terminal shall decline the transaction
  - If the CID indicated that the card returned a TC, the terminal shall not issue a second GENERATE AC command to complete the transaction.
  - If the CID indicated that the card returned an ARQC, the terminal shall issue a second GENERATE AC command requesting an AAC to complete the transaction. Although an XDA signature will be requested it shall not be verified by the terminal.

If the CID indicated that the card returned an AAC, the terminal shall decline the transaction without issuing a second GENERATE AC command.

6.3.3 Processing Restrictions

If the card and terminal Application Version Numbers are different, the terminal shall attempt to continue processing the transaction. If it is unable to continue, the terminal shall abort the transaction and should display the 'NOT ACCEPTED' message.

When processing the Application Usage Control, the terminal must know whether or not it is an ATM. See section A1 for information on identifying an ATM.

A terminal supporting cashback should not offer cashback facility to the cardholder if the Application Usage Control does not allow this option.

6.3.4 Cardholder Verification Processing

Recognition of a CVM means the CVM Code is understood by the terminal but not necessarily supported by the terminal. The terminal shall recognise the EMV-defined CVM codes in Book 3 section C3 including the CVM codes for 'No CVM required' and 'Fail CVM processing'. The terminal may also recognise proprietary CVM Codes not defined in section C3.

Support for a CVM means the terminal has the hardware and software necessary to perform the CVM. Support for EMV-defined CVM codes is indicated in the following manner:

- For EMV-defined CVM codes, support is indicated in Terminal Capabilities and Biometric Terminal Capabilities.

[2] An XDA signature is requested only for the sake of consistency – the terminal ignores the XDA signature provided in the response to the second GENERATE AC because it is known that if the issuer has authorised the transaction then it was despite XDA failure on the first GENERATE AC.

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- For CVM codes not defined in EMV, support may be known implicitly.
- For Combination CVMs, both CVM codes must be supported.
- Fail CVM shall always be considered supported.

A CVM can be performed only if it is both recognised and supported.

6.3.4.1 Offline CVM

When the applicable CVM is an offline PIN, the terminal should issue a GET DATA command to the card to retrieve the PIN Try Counter prior to issuing either the VERIFY command or GET CHALLENGE command.

If the PIN Try Counter is not retrievable or the GET DATA command is not supported by the ICC, or if the value of the PIN Try Counter is not zero, indicating remaining PIN tries, the terminal shall prompt for PIN entry such as by displaying the message 'ENTER PIN'.

If the value of the PIN Try Counter is zero, indicating no remaining PIN tries, the terminal should not allow offline PIN entry. The terminal:

- shall set the 'PIN Try Limit exceeded' bit in the TVR to 1 (for details on TVR, see Book 3 Annex C),
- shall not display any specific message regarding PINs, and
- shall continue cardholder verification processing in accordance with the card's CVM List.

If offline PIN verification by the ICC is successful, the terminal shall set byte 3 of the CVM Results to 'successful'. Otherwise, the terminal shall continue cardholder verification processing in accordance with the card's CVM List. (CVM Results is described in section 6.3.4.5 and coded according to section A4.)

6.3.4.2 Online CVM

When the applicable CVM is an online PIN, the IFD shall not issue a VERIFY command. Instead, the PIN pad shall encipher the PIN upon entry for transmission in the authorisation or financial transaction request.

The terminal shall allow a PIN to be entered for online verification even if the card's PIN Try Limit is exceeded.

The terminal shall set byte 3 of the CVM Results to 'unknown'.

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6.3.4.3 PIN Entry Bypass

If a PIN is required for entry as indicated in the card's CVM List, an attended terminal with an operational PIN pad may have the capability to bypass PIN entry before or after several unsuccessful PIN tries.[3] If this occurs, the terminal:

- shall set the 'PIN entry required, PIN pad present, but PIN was not entered' bit in the TVR to 1,
- shall not set the 'PIN Try Limit exceeded' bit in the TVR to 1,
- shall consider this CVM unsuccessful, and
- shall continue cardholder verification processing in accordance with the card's CVM List.

When PIN entry has been bypassed for one PIN-related CVM, it may be considered bypassed for any subsequent PIN-related CVM during the current transaction.

6.3.4.4 Signature

When the applicable CVM is signature, the terminal shall set byte 3 of the CVM Results to 'unknown'. At the end of the transaction, the terminal shall provide a way to capture cardholder signature (e.g., by printing a line for the cardholder signature on the merchant's copy of the receipt, or by using an electronic panel). (See section A2 for requirements for the terminal to support signature as a CVM.)

6.3.4.5 CVM Results

When the applicable CVM is 'No CVM required', if the terminal supports 'No CVM required' it shall set byte 3 of the CVM Results to 'successful'. When the applicable CVM is 'Fail CVM processing', the terminal shall set byte 3 of the CVM Results to 'failed'.

The terminal shall set bytes 1 and 2 of the CVM Results with the Method Code and Condition Code of the last CVM performed. After a successful CVM, CVM Results reflect the successful CVM. After an unsuccessful CVM with byte 1 bit 7 of the CV Rule set to 0 (fail cardholder verification), CVM Results reflect the unsuccessful CVM. After an unsuccessful CVM with byte 1 bit 7 set to 1 (apply succeeding CVM), CVM Results reflect this unsuccessful CVM only when no subsequent CVMs are performed; that is, when for each subsequent CV Rule either the CVM condition is not satisfied or the CVM condition is satisfied but the CVM method is not supported or is not recognised. If a subsequent CVM is performed, CVM Results reflect the outcome of the subsequent CVM.

If the last CVM performed was not considered successful (byte 3 of the CVM Results is not set to 'successful' or 'unknown'), the terminal shall set byte 3 of the CVM Results to 'failed'.

If no CVM was performed (no CVM List present or no CVM conditions satisfied), the terminal shall set byte 1 of the CVM Results to 'No CVM performed'.

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[3] This prevents a genuine cardholder who does not remember the PIN from having to keep entering incorrect PINs until the PIN is blocked in order to continue with the transaction.

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Table 2 on the following page shows the setting of CVM Results, Byte 3 of Terminal Verification Results (TVR) related to CVM processing, and the Transaction Status Information (TSI) bit for CVM processing for some conditions that may occur during CVM List processing. Please note that for the first five entries in the table, the second byte of the CVM Results has no actual meaning so it is recommended it be set to '00'.

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Conditions	Corresponding CVM Results			TVR Byte 3	TSI Byte 1 bit 7
	Byte 1 (CVM Performed)	Byte 2 (CVM Condition)	Byte 3 (CVM Result)		
When the card does not support cardholder verification (AIP bit 5=0)	'3F' (Book 4 section 6.3.4.5 and section A4)	'00' (no meaning)	'00'	--	0
When CVM List is not present or the CVM List has no CVM rules	'3F' (Book 4 section 6.3.4.5 and section A4)	'00' (no meaning)	'00'	--	0 (Book 3 section 10.5)
When no CVM Conditions in CVM List are satisfied	'3F' (Book 4 section 6.3.4.5 and section A4)	'00' (no meaning)	'01'	bit 8 = 1	1
When the terminal does not support any of the CVM Codes in CVM List where the CVM Condition was satisfied	'3F' (Book 4 section 6.3.4.5 and section A4)	'00' (no meaning)	'01'	bit 8 = 1	1
When the terminal does not recognise any of the CVM Codes in CVM List (or the code is RFU) where the CVM Condition was satisfied	'3F' (Book 4 section 6.3.4.5 and section A4)	'00' (no meaning)	'01'	bit 8 = 1 bit 7 = 1	1
When the (last) performed CVM is 'Fail CVM processing'	'00' or '40' (Not recommended for cards.)	The CVM Condition Code for the CVM Code in Byte 1	'01' (Book 4 section 6.3.4.5 and section A4)	bit 8 = 1	1
When the last performed CVM fails	The CVM Code for the last performed CVM	The CVM Condition Code for the CVM Code in Byte 1	'01'	bit 8 = 1[4]	1
When the CVM performed is 'Signature'	'1E' or '5E' (CVM Code for 'Signature')	The CVM Condition Code for the CVM Code in Byte 1	'00' (Book 4 section 6.3.4.4 and section A4)	--	1

Table 2: Setting of CVM Results, TVR bits, and TSI bits following CVM Processing

[4] Errors with PIN related CVMs may also result in the setting of other PIN-related TVR bits.

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Conditions	Corresponding CVM Results			TVR Byte 3	TSI Byte 1 bit 7
	Byte 1 (CVM Performed)	Byte 2 (CVM Condition)	Byte 3 (CVM Result)		
When the CVM performed is 'Online PIN'	'02' or '42' (CVM Code for 'Online PIN')	The CVM Condition Code for the CVM Code in Byte 1	'00' (Book 4 section 6.3.4.2 and section A4)	bit 3 = 1	1
When the CVM performed is 'No CVM required'	'1F' or '5F' (CVM code for 'No CVM Required')	The CVM Condition Code for the CVM Code in Byte 1	'02' (Book 4 section 6.3.4.5 and section A4)	--	1
When a CVM is performed and fails with the failure action being 'Go to Next' and then the end of the CVM List is reached without another CVM Condition being satisfied	The last CVM Code in the CVM List for which the CVM Condition was satisfied (but that failed).	The CVM Condition Code for the CVM Code in Byte 1	'01'	bit 8 = 1	1
When the CVM performed is a combination CVM Code, and at least one fails and the failure action is Fail CVM	CVM Code for the combination CVM	The CVM Condition Code for the CVM Code in Byte 1	'01'	bit 8 = 1[4]	1
When the CVM performed is a combination CVM Code, and one passes and the result of the other is 'unknown'	CVM Code for the combination CVM	The CVM Condition Code for the CVM Code in Byte 1	'00'	--	1

Table 2: Setting of CVM Results, TVR bits, and TSI bits following CVM Processing, continued

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The data elements Application Label, Application Preferred Name, Issuer Code Table Index, and Language Preference are present for the convenience of the cardholder and are not critical to the successful processing of application selection. If these data elements are present in the FCI, the issuer is responsible for their correct encoding.

If the Application Label data element is not present in the FCI of an ADF, the terminal shall not terminate the card session but shall proceed with application selection.

Terminals shall not enforce the correct formatting of these data elements. If Application Preferred Name or Application Label contains a character that is not valid for the defined format, the terminal shall display the character if it is able to, or if the terminal is unable to display the invalid character, it should omit the character or substitute a space character or any other appropriate character. Otherwise, if the terminal detects format errors in any of these data elements, the terminal shall disregard these errors and act as if the response provided by the card did not contain these data elements. More specifically, the terminal shall not terminate the card session but shall proceed with application selection.

If the terminal does not understand the value in Issuer Code Table Index or Language Preference, it shall treat the data element as not present.

12.3 Building the Candidate List

A terminal shall always support application selection using the List of AIDs method as described in section 12.3.3. A terminal may additionally support application selection using the PSE method as described in section 12.3.2. If the card contains no PSE, the procedure described in section 12.3.3 must be followed.

The terminal may know other ways that are not described in this section to locate proprietary applications. This is permitted as long as all interoperable applications can be located in the ICC using the techniques described here.

If allowed by business agreement between the affected parties, specific applications may be eliminated from consideration either during or after building the candidate list.

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The terminal shall maintain a list of applications supported by the terminal identified by their AIDs. The terminal determines which applications in the ICC are supported by comparing the AIDs for applications supported by the terminal with the DF Names[3] of applications supported by the ICC.

For each of the AIDs within the list of applications supported by the terminal, the terminal shall use the Application Selection Indicator (ASI) as an indicator of the matching criterion to use.

- If the ASI indicates that the terminal supports the ICC application only when the AID in the terminal has the same length and value as the DF Name then this limits the ICC to at most one matching ADF.
- If the ASI indicates that the terminal supports the ICC application when the DF Name begins with the entire AID kept within the terminal then this allows the ICC to have multiple ADFs matching the terminal AID by adding unique information to the DF Name used by each of the ADFs.

If the ICC does not support partial name selection as described in section 11.3.5, the DF Name of the ADF must be an exact match with the terminal AID.

If the ICC supports partial name selection as described in section 11.3.5 and has multiple ADFs supported by a single terminal AID, all of the matching DF Names must be distinguished by adding unique data to the PIX. All of the matching DF Names shall be longer than the corresponding terminal AID.

[3] Depending upon the method used to build the candidate list, the names in the list will be ADF Names found in directory entries if the PSE selection method is used or DF Names found in the FCIs returned to SELECT commands if the List of AIDs method is used. For readability in this section, the term DF Name is used in place of either.

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If a terminal chooses to support application selection using the PSE method, it shall follow the procedure described in this section to determine the applications supported by the card. Figure 3 is a flow diagram of the logic described here.

The terminal performs the following steps:

1. The terminal begins by selecting the PSE using a SELECT command as described in section 11 using a file name of '1PAY.SYS.DDF01'. This establishes the PSE and makes the Payment System Directory accessible.

   If the card is blocked or the SELECT command is not supported (both conditions represented by SW1 SW2 = '6A81'), the terminal terminates the session.

   If there is no PSE in the ICC, the ICC shall return '6A82' ('File not found') in response to the SELECT command for the PSE. In this case, the terminal shall use the List of AIDs method described in section 12.3.3.

   If the PSE is blocked, the ICC shall return '6283'. In this case, the terminal shall use the List of AIDs method described in section 12.3.3.

   If the ICC returns SW1 SW2 = '9000', the terminal proceeds to step 2.

   If the card returns any other value in SW1 SW2, the terminal shall use the List of AIDs method described in section 12.3.3.

   If any error, including a SW1 SW2 different from '90 00' or '6A 83', occurs in steps 2 through 4, the terminal shall clear the candidate list and restart the application selection process using the List of AIDs method described in section 12.3.3 to find the matching applications.

2. The terminal uses the Directory SFI from the FCI returned and reads all the records in the Payment System Directory beginning with record number 1 and continuing with successive records until the card returns SW1 SW2 = '6A83', which indicates that the record number requested does not exist. (The card shall return '6A83' if the record number in the READ RECORD command is greater than the number of the last record in the file). If the card returns SW1 SW2 = '6A83' in response to a READ RECORD for record number 1 for the Payment System Directory, no directory entries exist, and step 5 (below) applies.

   For each record in the Payment System Directory, the terminal begins with the first directory entry and processes each directory entry in turn as described in steps 3 and 4. If there are no directory entries in the record, the terminal proceeds to the next directory record.

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the terminal as defined in section 12.3.1, the application joins the candidate list for final application selection under control of the ASI maintained in the terminal for that AID.

The ASI indicates whether the AID in the terminal shall match exactly (both in length and name) or need only partially match the associated ADF Name in the directory entry (tag '4F').

The application is added to the candidate list in either of the following cases:

- the ADF Name in the directory entry retrieved is an exact match, or
- the ASI for the AID in the terminal indicates that a partial match is allowed.

The application is not added to the candidate list if the ADF Name in the directory entry retrieved is not an exact match and the ASI for the AID in the terminal indicates that an exact match is required.

4. When the terminal finishes processing all entries in the last record of the Payment System Directory, all ADFs that can be found by this procedure have been determined. The search and the candidate list are complete. If at least one matching ADF Name was found, the terminal continues processing as described in section 12.4.
5. If steps 1 through 4 yield no directory entries that match applications supported by the terminal, the terminal shall use the list of AIDs method described in section 12.3.3 to find a match.

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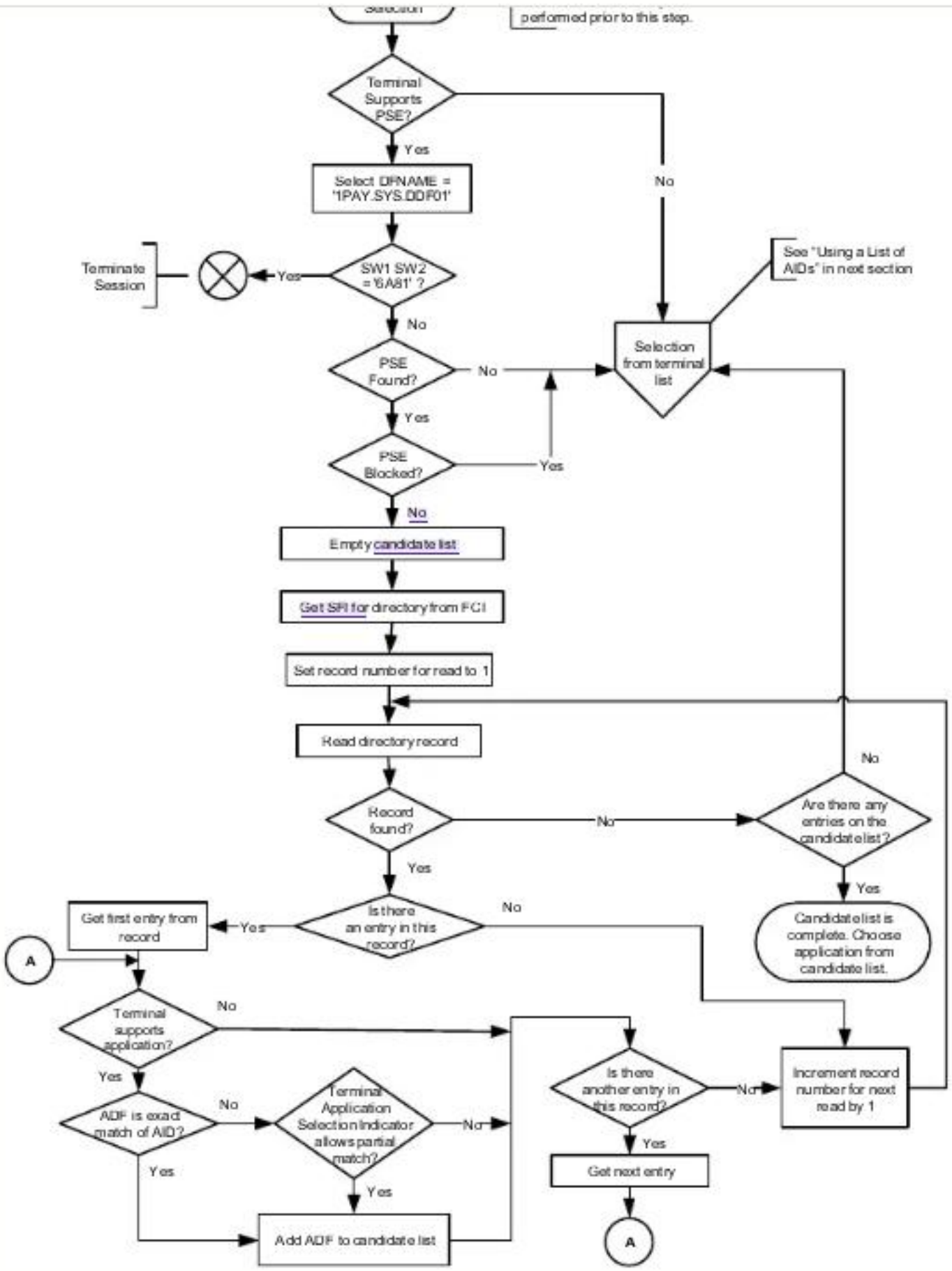


Figure 3: Terminal Logic Using Directories

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If either the card or the terminal does not support the PSE method or if the terminal is unable to find a matching application using the Payment System Directory selection method, the terminal shall use a list of AIDs that it supports to build the candidate list. Figure 4 is a flow diagram of the logic described here.

The terminal performs the following steps:

1. The terminal issues a SELECT command using the first AID [4] in the terminal list as the file name.
2. If the SELECT command fails because the card is blocked or the command is not supported by the ICC (SW1 SW2 = '6A81'), the terminal terminates the card session.
3. If the SELECT command is successful (SW1 SW2 = '9000' or '6283'), the terminal compares the AID with the DF Name field returned in the FCI. The DF Name will either be identical to the AID (including the length), or the DF Name will start with the AID but will be longer. If the names are identical, the terminal proceeds with step 4. If the DF Name is longer, the card processed the command as a partial name selection, and the terminal proceeds to step 6.

   If the card returns any other status or if mandatory data is missing from the SELECT response or if the FCI contains formatting errors not described in Section 12.2.4, the terminal proceeds to Step 5 without adding the DF Name to the candidate list.
4. If the SELECT command is successful (SW1 SW2 = '9000'), the terminal adds the DF Name and other information [5] from the FCI to the candidate list and proceeds to step 5. If the application is blocked (SW1 SW2 = '6283'), the terminal proceeds to step 5 without adding the DF Name to the candidate list.
5. The terminal issues another SELECT command using the next AID in its list and returns to step 3. If there are no more AIDs in the list, the candidate list is complete, and the terminal proceeds as specified in section 12.4.
6. Along with the AID list, the terminal keeps an Application Selection Indicator that indicates whether the card may have multiple occurrences of the application within the card. The terminal checks this indicator. If the indicator says that an exact match (in both length and name) is required, the terminal does not add the DF Name and other information from the FCI to the candidate list, but proceeds to step 5.

---

[4] To assist in a clear understanding of the process described in this section, it is necessary to distinguish between the application identifier kept in the terminal and the application identifier kept in the ICC. As can be seen in section 12.3.1, these might not be identical even for matching applications. In this procedure, the term AID is used for the application identifier kept in the terminal, and DF Name is used for the application identifier in the card.

[5] The Application Label and Application Preferred Name must also be saved if the cardholder will be provided a list during final selection. The DF Name and the Application Priority Indicator will be required in any case.

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application is not blocked (SW1 SW2 = '9000'), the terminal adds the DF Name and other information from the FCI to the candidate list and proceeds to step 7.

If multiple occurrences are permitted but the application is blocked (SW1 SW2 ≠ '9000'), the terminal proceeds to step 7 without adding the DF Name or other information from the FCI to the candidate list.

7. The terminal repeats the SELECT command using the same command data as before, but changes P2 in the command to '02' (Select Next). If the ICC returns SW1 SW2 = '9000', '62xx', or '63xx', the terminal returns to step 3. If it returns a different SW1 SW2, the terminal goes to step 5.

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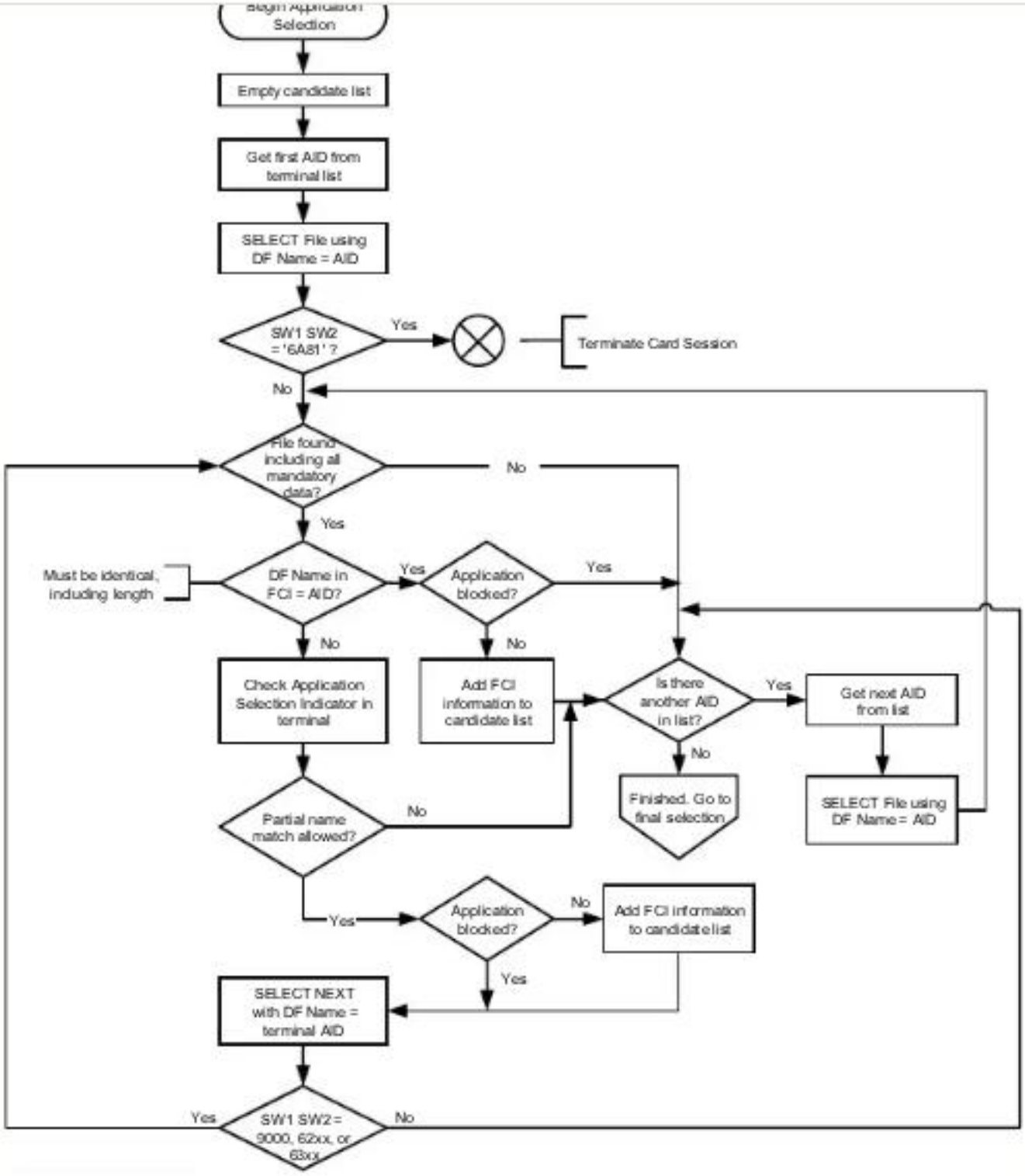


Figure 4: Using the List of AIDs in the Terminal

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The terminal should support the ability to allow the cardholder to select an application or to confirm the application proposed by the terminal.

- Cardholder selection is a function that allows the cardholder to choose the application they want to use when two or more applications are mutually supported by the card and terminal.
- Cardholder confirmation is a mechanism that allows issuers/cardholders the ability to identify applications on the card that the terminal shall not be allowed to select automatically without explicit cardholder confirmation of some kind.

These two features are covered by a single implementation or configuration option. These specifications allow for the following:

- The terminal supports both cardholder selection and cardholder confirmation.
- The terminal supports neither cardholder selection nor cardholder confirmation.

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- the Application Preferred Name, if present and if the Issuer Code Table Index indicating the part of ISO/IEC 8859 to use is present and supported by the terminal (as indicated in Additional Terminal Capabilities)
- otherwise the Application Label, if present, by using the common character set of ISO/IEC 8859 (see Book 4 Annex B)

Once the terminal determines the list of mutually supported applications, it proceeds as follows:

1. If there are no mutually supported applications, the transaction is terminated.
2. If there is only one mutually supported application, the terminal checks b8 of the card's Application Priority Indicator for that application if present.
    - If b8 = 0, the terminal selects the application.
    - If b8 = 1 and the terminal provides for confirmation by the cardholder, the terminal requests confirmation and selects the application if the cardholder approves. If the terminal does not provide for confirmation by the cardholder, or if the terminal requests confirmation and the cardholder does not approve, the terminal terminates the session.
3. If there is more than one mutually supported application, then:
    - If the terminal supports the ability to allow the cardholder to select an application the terminal shall offer a selection to the cardholder as described in step 4.
    - If the terminal does not support the ability to allow the cardholder to select an application, the terminal makes the selection itself as described in step 5.

    Step 4 is the preferred method.
4. When the terminal offers a selection to the cardholder, then:
    - Applications where the card's Application Priority Indicator is present shall be presented in priority sequence as indicated by the Application Priority Indicator with the highest priority application offered first. Where the same priority is assigned to multiple applications, the terminal may present these applications in its own preferred order or in the order encountered in the card.
    - For applications where the card's Application Priority Indicator is not present, the terminal may present these applications in its own preferred order or in the order encountered in the card.
5. When the terminal does not offer a selection to the cardholder, then the terminal shall select the highest priority application that does not require cardholder confirmation (that is, the card's Application Priority Indicator b8 = 0) from the list of mutually supported applications. The application's priority is determined using the same rules as described in step 4.

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application shall be selected. A SELECT command coded according to section 11 shall be issued by the terminal for the application using the ADF Name field (if a directory was read) or the DF Name field from the FCI (if the List of AIDs method was used) found during the building of the candidate list.

On successful selection of the application to be used (SW1 SW2 = '9000' returned in response to the final SELECT command, the response contains no format errors other than those described in section 12.2.4, and the AID used in the final SELECT command exactly matches the DF Name (tag '84') returned by the ICC in the FCI), the terminal shall set the value of the terminal data element Application Identifier (AID) – terminal (tag '9F06') to the same value as the DF Name (tag '84') returned in the FCI. If transaction processing is to be continued according to Book 3, this shall be done prior to issuance of the GET PROCESSING OPTIONS command.

If the command returns other than '9000' in SW1 SW2 or the SELECT response contains format errors other than those described in section 12.2.4, the application shall be removed from the candidate list, and processing shall resume at step 1.

If the cardholder selects or confirms the selection of an application that is subsequently removed from the candidate list due to its being blocked or for any other reason, no application is to be selected without cardholder confirmation.

If no application can be selected, the terminal shall terminate the transaction.

In any case, the terminal shall inform the cardholder of the action taken (that is, by using the messages defined in Book 4 section 11.2), if appropriate.

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Usage of the Application Selection Registered Proprietary Data (ASRPD) received from the ICC is optional and proprietary.

If Application Selection processing does not use ASRPD,
then Application Selection processing shall ignore instances of '9F0A' in Directory Entries (Tag '61') and continue processing as if the data was not present.

If Application Selection processing uses ASRPD,
then Application Selection processing interprets the value field to recover all the Proprietary Data Identifiers:

- **If** the value field of the ASRPD is not correctly formatted (ID L V, ID L V, ... as defined below),
  then the Application Selection processing shall ignore this instance of the ASRPD and continue processing as if the data was not present. Note that no assumption can be made on the IDs already registered by EMVCo nor on the format of the value fields of the Proprietary Data Identifiers and as a consequence the value field of the ASRPD is considered to be incorrectly formatted only if a length problem is detected.
- **If** the value field of the ASRPD is correctly formatted,
  then proprietary functionality may be activated for the recognised Proprietary Data Identifiers.
- Application Selection processing is not required to keep track of the Proprietary Data Identifiers defined by EMVCo, therefore unrecognised Proprietary Data Identifiers shall be ignored.

The coding of the ASRPD is as follows:

The value field of the ASRPD object follows the following format:
ID1, L1, V1, ID2, L2, V2,...

Where

- ID is a two byte Proprietary Data Identifier. Proprietary Data Identifiers are registered by EMVCo.
- L is the length of the value field coded in 1 byte (0 to 255).
- V is the value field. Its content is proprietary and format is out of scope of EMVCo.

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Part IV

Annexes

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Annex A Removed in Version 4.4

Annex A was deleted when Part II was removed in Version 4.4.

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Annex B Data Elements Table

Table 14 defines those data elements that may be used for application selection and their mapping onto data objects and files.[6] Table 15 lists the data elements in tag sequence.

The characters used in the "Format" column are described in section 4.3, Data Element Format Convention.

B1 Data Elements by Name

Table 14: Data Elements Table

Name	Description		Source	Format	Template	Tag	Length
Application Dedicated File (ADF) Name	Identifies the application as described in ISO/IEC 7816-4		ICC	b	'61'	'4F'	5–16
Application Identifier (AID) - terminal	Identifies the application as described in ISO/IEC 7816-4		Terminal	b	—	'9F06'	5–16

[6] Book 3 Annex A provides a complete data elements table, defining all data elements that may be used for financial transaction interchange and their mapping onto data objects and files.

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Name	Description		Source	Format	Template	Tag	Length
Application Label	Mnemonic associated with the AID according to ISO/IEC 7816-4		ICC	ans with the special character limited to space	'61' or 'A5'	'50'	1–16
Application Preferred Name	Preferred mnemonic associated with the AID		ICC	ans (see section 4.3)	'61' or 'A5'	'9F12'	1–16
Application Priority Indicator	Indicates the priority of a given application or group of applications in a directory		ICC	b	'61' or 'A5'	'87'	1
Application Selection Indicator	For an application in the ICC to be supported by an application in the terminal, the Application Selection Indicator indicates whether the associated AID in the terminal must match the AID in the card exactly, including the length of the AID, or only up to the length of the AID in the terminal There is only one Application Selection Indicator per AID supported by the terminal		Terminal	At the discretion of the terminal. The data is not sent across the interface	—	—	See Format

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Name	Description		Source	Format	Template	Tag	Length
Application Selection Registered Proprietary Data (ASRPD)	Proprietary data allowing for proprietary processing during application selection. Proprietary data is identified using Proprietary Data Identifiers that are managed by EMVCo and their usage by the Application Selection processing is according to their intended usage, as agreed by EMVCo during registration		Card	b, also see section 12.5	'73' 'BF0C'	'9F0A'	var.
Application Template	Contains one or more data objects relevant to an application directory entry according to ISO/IEC 7816-4		ICC	b	'70'	'61'	var. up to 252
Bank Identifier Code (BIC)	Uniquely identifies a bank as defined in ISO 9362.		ICC	var.	'BF0C' or '73'	'5F54'	8 or 11
Dedicated File (DF) Name	Identifies the name of the DF as described in ISO/IEC 7816-4		ICC	b	'6F'	'84'	5–16
Directory Definition File (DDF) Name	Identifies the name of a DF associated with a directory		ICC	b	'61'	'9D'	5–16
Directory Discretionary Template	Issuer discretionary part of the directory according to ISO/IEC 7816-4		ICC	var.	'61'	'73'	var. up to 252
File Control Information (FCI) Issuer Discretionary Data	Issuer discretionary part of the FCI		ICC	var.	'A5'	'BF0C'	var. up to 222

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Name	Description		Source	Format	Template	Tag	Length
File Control Information (FCI) Proprietary Template	Identifies the data object proprietary to this specification in the FCI template according to ISO/IEC 7816-4		ICC	var.	'6F'	'A5'	var.
File Control Information (FCI) Template	Identifies the FCI template according to ISO/IEC 7816-4		ICC	var.	—	'6F'	var. up to 252
International Bank Account Number (IBAN)	Uniquely identifies the account of a customer at a financial institution as defined in ISO 13616.		ICC	var.	'BF0C' or '73'	'5F53'	Var. up to 34
Issuer Code Table Index	Indicates the code table according to ISO/IEC 8859 for displaying the Application Preferred Name		ICC	n 2	'A5'	'9F11'	1
Issuer Country Code (alpha2 format)	Indicates the country of the issuer as defined in ISO 3166 (using a 2 character alphabetic code)		ICC	a 2	'BF0C' or '73'	'5F55'	2
Issuer Country Code (alpha3 format)	Indicates the country of the issuer as defined in ISO 3166 (using a 3 character alphabetic code)		ICC	a 3	'BF0C' or '73'	'5F56'	3
Issuer Identification Number (IIN)	The number that identifies the major industry and the card issuer and that forms the first part of the Primary Account Number (PAN)		ICC	n 6	'BF0C' or '73'	'42'	3

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Name	Description		Source	Format	Template	Tag	Length
Issuer Identification Number Extended (IINE)	The number that identifies the major industry and the card issuer and that forms the first part (6 or 8-digits) of the Primary Account Number (PAN). While the first 6-digits of the IINE (tag '9F0C') and IIN (tag '42') are the same and there is no need to have both data objects on the card, cards may have both the IIN and IINE data objects present.		ICC	n 6 or 8	'BF0C' or '73'	'9F0C'	var. 3 or 4
Issuer URL	The URL provides the location of the issuer's Library Server on the Internet		ICC	ans	'BF0C' or '73'	'5F50'	var.
Language Preference	1–4 languages stored in order of preference, each represented by 2 alphabetical characters according to ISO 639 **Note:** EMVCo strongly recommends that cards be personalised with data element '5F2D' coded in lowercase, but that terminals accept the data element whether it is coded in upper or lower case.		ICC	an 2	'A5'	'5F2D'	2–8
Log Entry	Provides the SFI of the Transaction Log file and its number of records		ICC	b	'BF0C' or '73'	'9F4D'	2

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Name	Description		Source	Format	Template	Tag	Length
Processing Options Data Object List (PDOL)	Contains a list of terminal resident data objects (tags and lengths) needed by the ICC in processing the GET PROCESSING OPTIONS command		ICC	b	'A5'	'9F38'	var.
READ RECORD Response Message Template	Contains the contents of the record read. (Mandatory for SFIs 1-10. Response messages for SFIs 11-30 are outside the scope of EMV, but may use template '70'.)		ICC	var.	—	'70'	var. up to 252
Short File Identifier (SFI)	Identifies the AEF referenced in commands related to a given ADF or DDF. It is a binary data object having a value in the range 1 – 30 and with the three high order bits set to zero.		ICC	b	'A5'	'88'	1

When the length defined for the data object is greater than the length of the actual data, the following rules apply:

- A data element in format n is right justified and padded with leading hexadecimal zeroes.
- A data element in format a, an, or ans is left justified and padded with trailing hexadecimal zeroes.

When data is moved from one entity to another (for example, card to terminal), it shall always be passed in order from high order to low order, regardless of how it is internally stored. The same rule applies when concatenating data.

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B2 Data Elements by Tag

Name	Template	Tag
Issuer Identification Number (IIN)	'BF0C' or '73'	'42'
Application Dedicated File (ADF) Name	'61'	'4F'
Application Label	'61' or 'A5'	'50'
Language Preference	'A5'	'5F2D'
Issuer URL	'BF0C' or '73'	'5F50'
International Bank Account Number (IBAN)	'BF0C' or '73'	'5F53'
Bank Identifier Code (BIC)	'BF0C' or '73'	'5F54'
Issuer Country Code (alpha2 format)	'BF0C' or '73'	'5F55'
Issuer Country Code (alpha3 format)	'BF0C' or '73'	'5F56'
Application Template	'70'	'61'
File Control Information (FCI) Template	—	'6F'
READ RECORD Response Message Template	—	'70'
Directory Discretionary Template	'61'	'73'
Dedicated File (DF) Name	'6F'	'84'
Application Priority Indicator	'61' or 'A5'	'87'
Short File Identifier (SFI)	'A5'	'88'
Directory Definition File (DDF) Name	'61'	'9D'
Application Identifier (AID) - terminal	—	'9F06'
Application Selection Registered Proprietary Data (ASRPD)	'73'	'9F0A'
Issuer Identification Number Extended (IINE)	'BF0C' or '73'	'9F0C'
Issuer Code Table Index	'A5'	'9F11'
Application Preferred Name	'61' or 'A5'	'9F12'
Processing Options Data Object List (PDOL)	'A5'	'9F38'
Log Entry	'BF0C' or '73'	'9F4D'
File Control Information (FCI) Proprietary Template	'6F'	'A5'
File Control Information (FCI) Issuer Discretionary Data	'A5'	'BF0C'

Table 15: Data Elements Tags

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Annex C Examples of Directory Structures

This annex illustrates some possible logical ICC file structures.

C1 Single Application Card

Figure 5 illustrates a single application card with only a single level directory. In this example, the MF (with file identification of '3F00', as defined by ISO/IEC 7816-4) acts as the only DDF in the card. The MF shall be given the unique payment system's name assigned to the first level DDF as defined in section 12.2, and the FCI of the MF shall contain the SFI data object.

'DIR A' in this example may or may not be the ISO DIR file, but it shall conform to this specification, including the requirement that it has a SFI in the range 1 to 10.

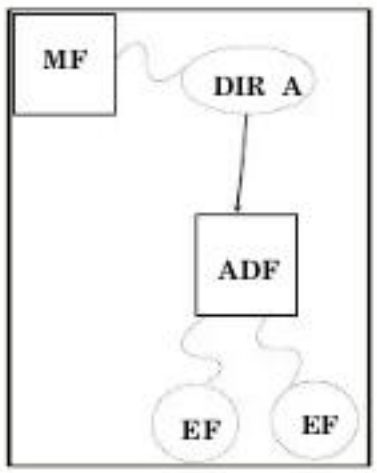


Figure 5: Simplest Card Structure Single Application

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Figure 6 gives an example of a multi-application card with a single directory. In this example, the root file (MF) does not support an application complying with this specification, and no restrictions are placed on the function of the MF. According to ISO/IEC 7816-4, a DIR file may be present but is not used by the application selection algorithm defined in section 12. Also note that the directory does not have entries for all ADFs (ADF2 to ADF5), as ADF5 is omitted. ADF5 can be selected only by a terminal that 'knows' ADF5 may exist in the card. The manner in which the terminal finds ADF5 is outside the scope of this specification.

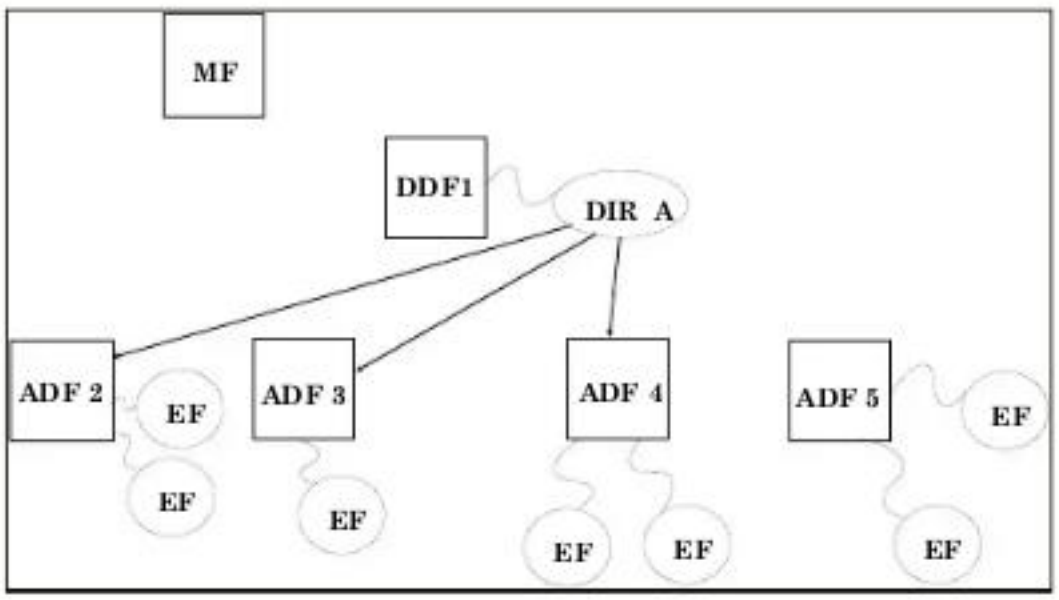


Figure 6: Single Level Directory

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Figure 7 is an example of a multi-application card with an *n* level directory structure. The first level directory ('DIR A') has entries for 2 ADFs – ADF3 and ADF4 – and a single DDF – DDF2. The directory attached to DDF2 ('DIR B') has entries for two ADFs – ADF21 and ADF22 – and a single DDF – DDF6. DDF5 has no entry in the root directory and can be found only by a terminal that 'knows' of the existence of DDF5. The manner in which the terminal finds and selects DDF5 is outside the scope of this specification, but the directory attached to DF5 ('DIR C') may conform to this specification, and, if found by the terminal, may lead the terminal to ADFs such as DF51, DF52, and DF53. DIR D, attached to DDF6, is a third level directory and points to four files (not shown), which may be either ADFs or more DDFs.

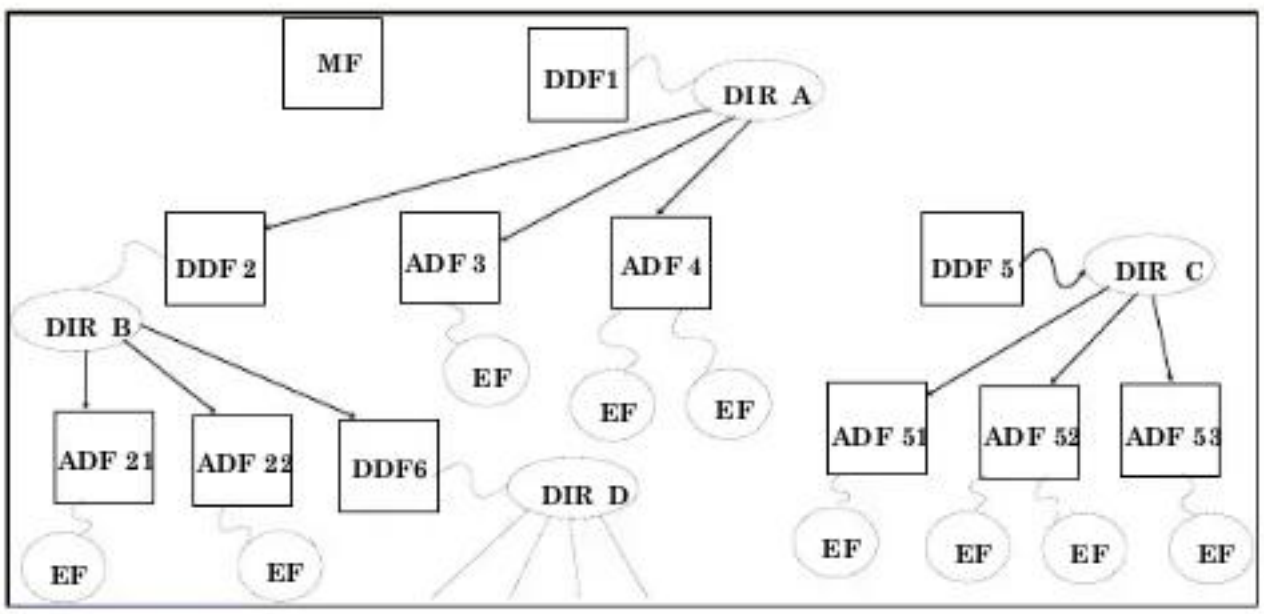


Figure 7: Third Level Directory

C4 Coding of Proprietary Directories

EMV does not mandate the formats for proprietary directories that may be present on the card in addition to the Payment System Directory. Directories following the structure defined in section 12.2 can be accessed using the methods described in this section.

These directories can have the same format as shown in Table 11 for the Payment System Directory Record Format, and can in addition to the one or more ADF Directory Entries also contain one or more DDF Directory Entries. If present, these DDF Directory entries can have the format as shown in Table 16 below.

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'9D'	5–16	DDF Name		M
'73'	var.	Directory Discretionary Template		O [7]
	'XXXX' (Tag constructed according to Book 3 Annex B)	var.	1 or more additional proprietary data elements from an application provider, issuer, or IC card supplier, or EMV-defined tags that are specifically allocated to template '73'	O

Table 16: Example of a DDF Directory Entry Format

[7] Other data objects not relevant to this specification may appear in this constructed data object.

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Part V

Common Core Definitions

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Common Core Definitions

This Part describes an optional extension to this Book, to be used when implementing the Common Core Definitions (CCD). It is to be used in conjunction with Books 2, 3, and 4, including the Common Core Definitions part of Books 2 and 3.

These Common Core Definitions specify a minimum common set of card application implementation options, card application behaviours, and data element definitions sufficient to accomplish an EMV transaction. Terminals certified to be compliant with the existing EMV specifications will, without change, accept cards implemented according to the Common Core Definitions, since the Common Core Definitions are supported within the existing EMV requirements. To be compliant with the Common Core Definitions, an implementation shall implement all the additional requirements in the Common Core Definitions sections of all affected Books.

Changed Sections

Each section heading below refers to the section in this Book to which the additional requirements apply. The text defines requirements for a common core implementation, in addition to the requirements already specified in the referenced section of EMV.

Part III - Files, Commands, and Application Selection

11 Commands

11.3 SELECT Command-Response APDUs

11.3.5 Processing State Returned in the Response Message

The ICC shall support partial name selection and shall accept SELECT command messages containing at least the 5 high-order bytes of the DF Name (that is, the RID). Select Next functionality shall be supported.

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